

$^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$ 2015MuZY

$J^\pi=0^+$ for ^{36}S ground state.

2015MuZY: A secondary beam of ^{36}S was produced via the projectile fragmentation of a 140-MeV/nucleon ^{48}Ca primary beam impinging on a 846-mg/cm² ^9Be target at the coupled cyclotron facility at NSCL, MSU. The ^{36}S nuclei were selected using the A1900 separator with an intensity of 8.1×10^5 pps and a purity of 89.7%. The excited states of ^{35}S were populated by the one-neutron knockout reaction from the ^{36}S beam at 88 MeV/nucleon (midtarget energy) on a 100-mg/cm² ^9Be secondary target located at the reaction target position of the S800 spectrometer. The projectile-like residues were identified from their energy loss measured by an ionization chamber located at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators situated at the object position and at the focal plane of the S800 spectrometer. Prompt γ rays from the deexcitation of the ^{35}S residues were detected by seven modules of the GRETINA Ge array. Measured E_γ , I_γ , (^{35}S) γ -coin, $\gamma\gamma$ -coin, the inclusive knockout cross section for producing ^{35}S from ^{36}S , the fractional populations and parallel momentum distributions of 11 populated states in ^{35}S residues. Deduced levels, J , π , L-transfers, partial knockout cross sections and spectroscopic factors. Calculated single-particle cross sections (σ_{sp}) for neutron removal and parallel momentum distributions using the eikonal model.

 ^{35}S Levels

b_f values under comments are population fraction in the knock-out reaction, σ_f^{sp} values are for calculated single-particle cross sections, and R_s values are quenching factors from a systematics study in [2014To14](#). The partial cross sections to each level can be deduced from $b_f \times \sigma_{\text{inc}}^{\text{exp}}$.

The inclusive cross section $\sigma_{\text{inc}}^{\text{exp}}$ for producing ^{35}S from ^{36}S is measured to be 63.7 mb 6.

$E(\text{level})^\dagger$	J^π	$L^\ddagger$	$C^2S^\text{@}$	Comments
0.0	$3/2^+$	2	3.2 7	$b_f=43.3\%$ 47, $\sigma_f^{\text{sp}}=13.1$ mb, $R_s=0.66$ 13.
1575 2	$1/2^+$	0	1.1 3	$b_f=18.6\%$ 31, $\sigma_f^{\text{sp}}=16.1$ mb, $R_s=0.64$ 13.
1991 2	$7/2^-$	3	<0.2	$b_f<2.9\%$, $\sigma_f^{\text{sp}}=12.7$ mb, $R_s=0.69$ 14.
2353 2	$3/2^-$	1	0.4 1	$b_f=5.3\%$ 13, $\sigma_f^{\text{sp}}=13.1$ mb, $R_s=0.70$ 14.
2720 2	$5/2^+$	2	0.6 2	$b_f=6.4\%$ 16, $\sigma_f^{\text{sp}}=11.4$ mb, $R_s=0.62$ 12.
2943 2		0	0.4 2	$b_f=5.7\%$ 16, $\sigma_f^{\text{sp}}=13.5$ mb, $R_s=0.61$ 12. L: inconsistent with $J^\pi=(3/2)^+$ in the Adopted Levels.
3425 2	$5/2^+$	2	0.5 1	$b_f=5.1\%$ 10, $\sigma_f^{\text{sp}}=11.0$ mb, $R_s=0.61$ 12.
3602 2		2	0.3 1	$b_f=3.0\%$ 7, $\sigma_f^{\text{sp}}=11.0$ mb, $R_s=0.60$ 12. L: inconsistent with $J^\pi=(7/2^+)$ in the Adopted Levels.
3821 3		3	0.3 1	$b_f=4.0\%$ 9, $\sigma_f^{\text{sp}}=11.8$ mb, $R_s=0.72$ 14. L: inconsistent with $J^\pi=(9/2^-)$ in the Adopted Levels.
3895 3		0	0.15 4	$b_f=2.3\%$ 5, $\sigma_f^{\text{sp}}=13.5$ mb, $R_s=0.72$ 14. L: inconsistent with $J^\pi=(5/2^+)$ in the Adopted Levels.
4028 2		2	0.6 1	$b_f=6.4\%$ 10, $\sigma_f^{\text{sp}}=12.1$ mb, $R_s=0.60$ 12.

[†] From a least-squares fit to γ -ray energies.

[‡] **2015MuZY** deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of residuals.

[#] As given in **2015MuZY**. Inconsistencies exist between the **2015MuZY** reported L values and J^π from other datasets.

[@] The spectroscopic factor $C^2S=b_f\sigma_{\text{inc}}^{\text{exp}}/R_s\sigma_f^{\text{sp}}$, where b_f is the population fraction, $\sigma_{\text{inc}}^{\text{exp}}$ is the measured total inclusive cross section, R_s is the quenching factor from a systematics study [2014To14](#), and σ_f^{sp} is the eikonal model calculated single-particle cross section.

 $\gamma(^{35}\text{S})$

$E_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π
778 2	2353	$3/2^-$	1575	$1/2^+$
1309 2	4028		2720	$5/2^+$
1542 2	3895		2353	$3/2^-$
1576 2	1575	$1/2^+$	0.0	$3/2^+$

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$^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$ **2015MuZY** (continued) $\gamma(^{35}\text{S})$ (continued)

E_γ [†]	$E_i(\text{level})$	J_i^π	E_f	J_f^π	E_γ [†]	$E_i(\text{level})$	J_i^π	E_f	J_f^π
1830 2	3821		1991	$7/2^-$	2720 2	2720	$5/2^+$	0.0	$3/2^+$
1990 [‡] 2	1991	$7/2^-$	0.0	$3/2^+$	2943 2	2943		0.0	$3/2^+$
2037 2	4028		1991	$7/2^-$	3425 2	3425	$5/2^+$	0.0	$3/2^+$
2352 2	2353	$3/2^-$	0.0	$3/2^+$	3602 2	3602		0.0	$3/2^+$

[†] From **2015MuZY**. Evaluators assume γ -ray energy uncertainties based on the statement in **2016Mu03** that γ -ray energy centroids were determined with an uncertainty of 2 keV.

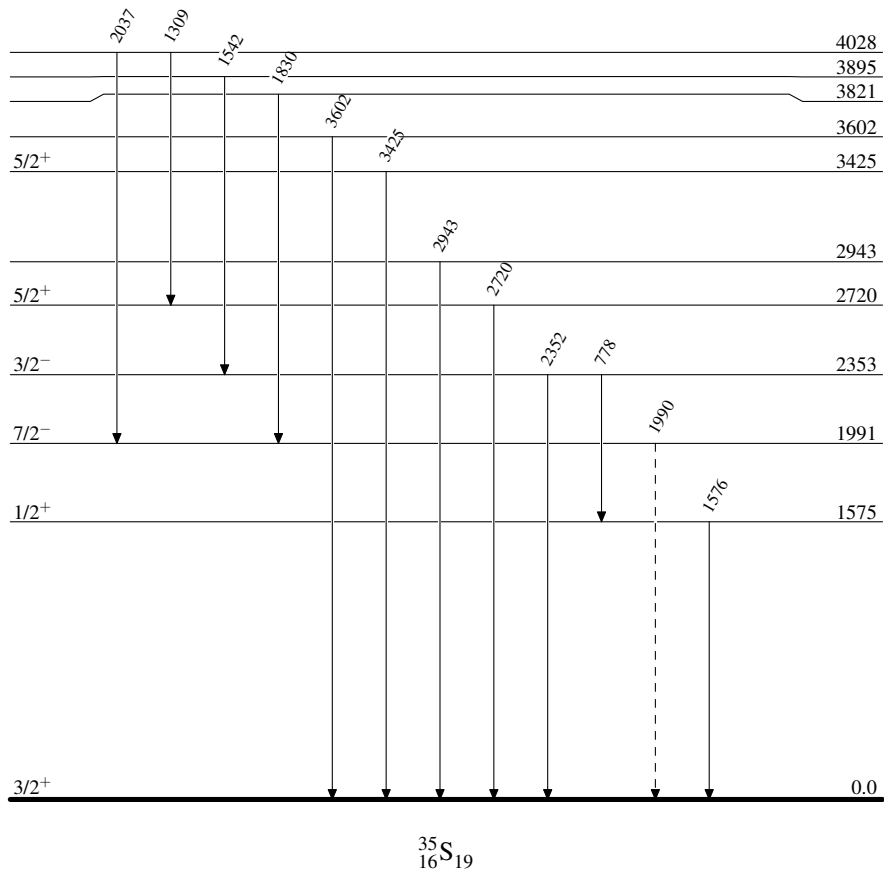
[‡] Placement of transition in the level scheme is uncertain.

⁹Be(³⁶S,³⁵Sγ) 2015MuZY

Legend

Level Scheme

-----▶ γ Decay (Uncertain)



³⁵S₁₉